

Macroeconomics II, Lecture IX:

Diamond-Mortensen-Pissarides: Statics

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- Last lecture: a framework for studying wage dispersion in a frictional labor market
- Today: a framework for studying unemployment in a frictional labor market
- In McCall, unemployment dynamics came from exogenous offer-arrival and separation rates
- Today we endogenize those rates by modeling vacancy creation and matching

The Diamond-Mortensen-Pissarides (DMP) model

- DMP emphasizes how jobs are endogenously created and destroyed
- Key references: Diamond (JPE 1981; JPE 1982; ReStud 1982), Mortensen (AER 1982; NBER 1982) and Pissarides (ReStud 1984; AER 1985)
- Sometimes referred to as the “Pissarides model” or the “Mortensen-Pissarides model”
- Large literature testing its assumptions/implications and proposing extensions

- ① Matching, market tightness, and the Beveridge curve
- ② Static DMP with exogenous separations
- ③ Extension: endogenous separations

The matching function

The job-finding rate and market tightness

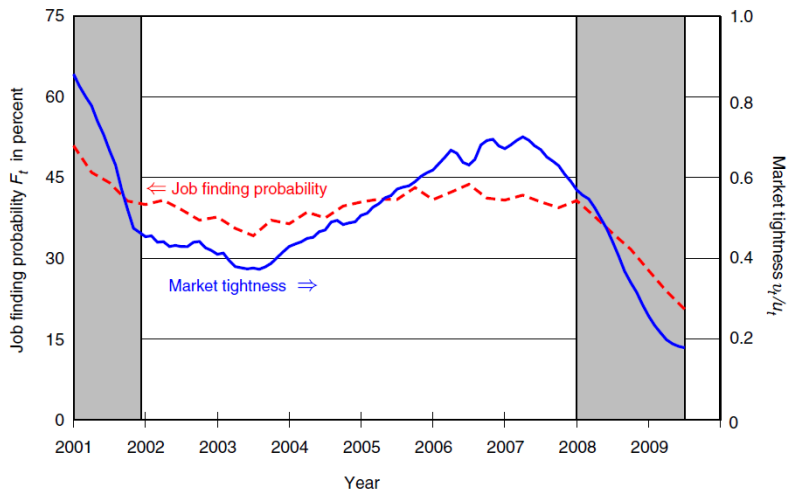
- Hypothesis: job search is a competitive process: it takes less time for a worker to find a job when demand is abundant and supply is scarce
- Supply = the number of unemployed U
- Demand = the number of vacancies V
- Define the *vacancy rate* as $v = \frac{V}{P}$, where P is the number of labor-force participants
- Define market tightness:

$$\theta \equiv \frac{V}{U} = \frac{v}{u}$$

► The market is tight when demand is high relative to supply

- **Testable implication:** Job-finding rate should be positively correlated with θ

Evidence on job finding and tightness



From Rogerson and Shimer (Handbook LE 2011).

The aggregate matching function

- The **aggregate matching function** endogenizes the job-finding rate via the competition analogy
- $M = M(U, V)$ gives the *flow rate of matches* as a function of the current level of unemployment and the number of vacant positions
- # matches in $\Delta t = M(U, V)\Delta t$
- $M(\cdot)$ should have two basic properties
 - ▶ increasing in both arguments
 - ▶ concave in both arguments
- Common assumption: $M(\cdot)$ is homogeneous of degree 1
 - ▶ Big gain in tractability
 - ▶ Support in aggregate time series data (Petrongolo-Pissarides, JEL 2001)
 - ▶ Without explicit microfoundations tied to micro evidence, it is difficult to assess this assumption
 - ★ Carrillo-Tudela, Gartner, and Kaas (2020) use German micro data; Skandalis (2019) uses French micro data. This remains an active research area.

Job-finding and job-filling rates

- With random matching, the aggregate matching function delivers a job-finding rate and a job-filling rate
- Homogeneity of degree 1 implies
 - ▶ rate at which unemployed worker meets vacant firms:
$$\lambda_u = \frac{M(U,V)}{U} = M(1, \frac{V}{U}) = M(1, \theta)$$
 - ▶ rate at which vacant firm meets unemployed workers:
$$\lambda_v = \frac{M(U,V)}{V} = M(\frac{U}{V}, 1) = M(\theta^{-1}, 1)$$
- $\lambda_u = \lambda_u(\theta)$ increasing and concave, $\lambda_v = \lambda_v(\theta)$ decreasing (and often convex)
- Note that $\lambda_u(\theta) = \theta \lambda_v(\theta)$

Example: Cobb-Douglas

- Common functional form: Cobb-Douglas $M(U, V) = AU^\alpha V^{1-\alpha}$
- Satisfies all assumptions: increasing, concave, homogeneous of degree 1
- Job-finding and job-filling rates:

$$\begin{aligned}\lambda_u(\theta) &= A\theta^{1-\alpha} \\ \lambda_v(\theta) &= A\theta^{-\alpha}\end{aligned}$$

- Parametric interpretation:
 - ▶ A : aggregate matching efficiency
 - ▶ α : elasticity of matches w.r.t. unemployment
- Note that $\log \lambda_u = \log A + (1 - \alpha) \log \theta$
 - ▶ Estimate A and α by regressing the (log) job-finding rate on (log) market tightness

Matching function implies a steady-state relationship between v and u

- Unemployment dynamics:

$$\dot{u} = \sigma(1 - u) - \lambda_u(\theta)u$$

- Steady state:

$$u = \frac{\sigma}{\sigma + \lambda_u(\theta)}$$

- With Cobb-Douglas:

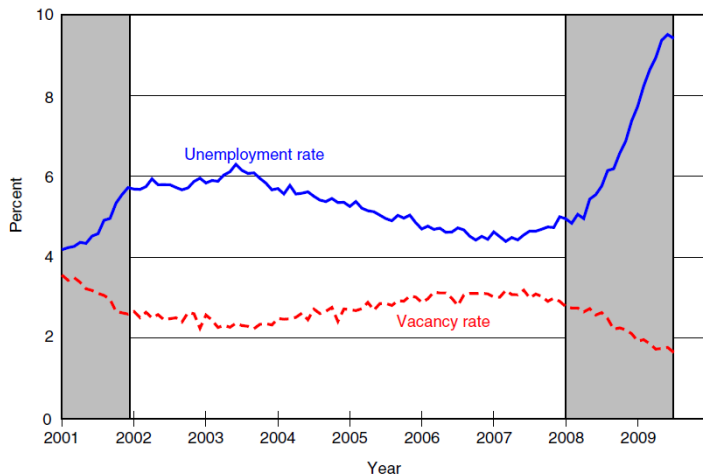
$$u = \frac{\sigma}{\sigma + A \left(\frac{v}{u}\right)^{1-\alpha}}$$

- Rewrite $v = f(u)$:

$$v = \left(\frac{\sigma}{A}\right)^{\frac{1}{1-\alpha}} \left(u^{-\alpha}(1 - u)\right)^{\frac{1}{1-\alpha}}$$

- Properties of f : for $0 < u < 1$, we have $f > 0$, $f' < 0$, and $f'' > 0$
 - ▶ $f' < 0 \Rightarrow \text{Corr}(u, v) < 0$

The Beveridge curve in the data



From Rogerson and Shimer (Handbook LE 2011).

- Negative correlation referred to as the **Beveridge curve**

Static equilibrium

- Continuous time; infinitely lived agents
- Workers are either employed or unemployed
- Firms are either non-operating, vacant, or filled
- A filled firm has one employee and produces flow output y
- Endogenous job-finding rate
- For now: Exogenous job-separation rate, endogenized later
- We abstract from heterogeneity, so there is no wage dispersion in this version
 - ▶ For the case with stochastic firm productivity, see Pissarides (AER 1985)
 - ▶ You can also integrate DMP with Burdett-Mortensen

Static DMP: three ingredients

- **Matching function**

- ▶ Increasing, concave, and homogeneous of degree 1
- ▶ Determines the job-finding and job-filling rates as functions of θ

- **Wage-setting rule**

- ▶ Our starting point: [Nash Bargaining](#)
- ▶ Because of matching frictions, a match creates a surplus
- ▶ Wages are set to split the match surplus between workers and firms
- ▶ Contrasts with wage posting, as in Burdett-Mortensen

- **Free entry**

- ▶ The vacancy market is competitive: the value of opening a vacancy must be 0 in equilibrium

- For now, focus on the static problem
- Workers take no decisions and there is no wage distribution
- The worker Bellman equations are

$$\begin{aligned}rW &= w + \sigma(U - W) \\ rU &= b + \lambda_u(\theta)(W - U)\end{aligned}$$

- Interpretation: flow value = flow payoff + arrival rate $\times$ capital gain or loss

- The firm's Bellman equations are

$$\begin{aligned}rJ &= (y - w) + \sigma(V - J) \\ rV &= -c + \lambda_v(\theta)(J - V)\end{aligned}$$

- c is the vacancy-posting cost and summarizes all hiring costs
- The firm's key margin is whether to post a vacancy
- Here is where the free-entry assumption comes into play
 - ▶ Firms open vacancies as long as $V \geq 0$
 - ▶ Competitive vacancy market, i.e., *free entry*: $V = 0$ in equilibrium

Wage setting

- The model has one price: the wage w
- Without further assumptions, the equilibrium wage is indeterminate
- Combining the worker Bellman equations, we get worker surplus:

$$\begin{aligned}r(W - U) &= w - b - (\sigma + \lambda_u(\theta))(W - U) \\ W - U &= \frac{w - b}{r + \sigma + \lambda_u(\theta)}\end{aligned}$$

- From the firm's Bellman equations and $V = 0$, we get firm surplus:

$$\begin{aligned}rJ &= (y - w) - \sigma J \\ J - V &= \frac{y - w}{r + \sigma}\end{aligned}$$

- For any wage $w \in [b, y]$, both the worker and the firm prefer a match to their outside option
- Standard closure: **Nash bargaining**

- When workers and firms meet, they sit down and bargain:

$$\max_w (W(w) - U)^\gamma (J(w) - V)^{1-\gamma}$$

- i.e., they maximize the geometric mean of their surpluses, with bargaining weights γ and $1 - \gamma$
- Nash (Econometrica 1950): the solution to this problem is also the unique solution to a general bargaining problem that satisfies four reasonable axioms
 - ▶ Notably, one axiom is Pareto efficiency
- It is by no means clear that this is a reasonable approximation of how wages are determined in the data, but it gives us a useful starting point

- Problem:

$$\max_w (W(w) - U)^\gamma (J(w) - V)^{1-\gamma}$$

- First order condition:

$$\gamma(J - V)W' + (1 - \gamma)(W - U)J' = 0$$

- From the definition of the value functions, we have $W'(w) = -J'(w)$ (why?)
- Hence

$$\gamma(J - V) = (1 - \gamma)(W - U)$$

Nash bargaining: an alternative expression

- Define total match surplus S :

$$S = (J - V) + (W - U)$$

- Combine with

$$\gamma(J - V) = (1 - \gamma)(W - U)$$

- to find

$$W - U = \gamma S$$

$$J - V = (1 - \gamma)S$$

- Nash bargaining: the worker and firm split the total surplus according to their bargaining power

Static equilibrium

- An equilibrium is a collection $\{W, U, J, V, w, \theta\}$ satisfying

- ▶ **Bellman equations:**

$$rW = w + \sigma(U - W)$$

$$rU = b + \lambda_u(\theta)(W - U)$$

$$rJ = (y - w) + \sigma(V - J)$$

$$rV = -c + \lambda_v(\theta)(J - V)$$

- ▶ **Free entry:**

$$V = 0$$

- ▶ **Wage setting rule:**

$$\gamma(J - V) = (1 - \gamma)(W - U)$$

- Six equations, six unknowns
- Given θ , unemployment and vacancies follow from

$$\dot{u} = \sigma(1 - u) - \lambda_u(\theta)u$$

$$u(0) = \underline{u}$$

$$\theta = \frac{v}{u}$$

Job creation curve

- Start from the firm side:

$$\begin{aligned}rJ &= (y - w) + \sigma (V - J) \\ rV &= -c + \lambda_v(\theta) (J - V)\end{aligned}$$

- Free entry in vacancy value equation implies:

$$J = \frac{c}{\lambda_v(\theta)}$$

- Free entry in job value equation implies:

$$J = \frac{y - w}{r + \sigma}$$

- Combining these gives

$$\frac{y - w}{r + \sigma} = \frac{c}{\lambda_v(\theta)} \quad \text{or} \quad w = y - \frac{c(r + \sigma)}{\lambda_v(\theta)}$$

- This is the **job-creation curve** in $\{w, \theta\}$ -space
 - Tells you how many vacancies per unemployed, θ , firms create given the wage w
 - Given $\lambda_v(\theta)$ decreasing, JC curve $w(\theta)$ is decreasing (and typically convex)

Wage curve

- Plug the worker and firm surplus expressions into the wage-setting rule

$$\begin{aligned}\gamma(J - V) &= (1 - \gamma)(W - U) \\ \gamma \left(\frac{y - w}{r + \sigma} \right) &= (1 - \gamma) \left(\frac{w - b}{r + \sigma + \lambda_u(\theta)} \right)\end{aligned}$$

- CRS matching: $\frac{\lambda_u(\theta)}{\lambda_v(\theta)} = \theta$ implies

$$\gamma \left(\frac{y - w}{r + \sigma} \right) = (1 - \gamma) \left(\frac{w - b}{r + \sigma + \theta \lambda_v(\theta)} \right)$$

- Use the job creation curve $y - w = \frac{c(r + \sigma)}{\lambda_v(\theta)}$ to substitute for $\lambda_v(\theta)$, eliminate terms and rearrange:

$$w = (1 - \gamma)b + \gamma(y + c\theta)$$

- This is the [wage curve](#)
- Together, WC and JC pin down θ and w

Steady-state unemployment

- Up to this point, we have not imposed a steady state anywhere in the solution
- In general, the level of unemployment and its dynamics is given by

$$\begin{aligned}\theta &= \frac{v}{u} \\ \dot{u} &= \sigma(1-u) - \lambda_u(\theta)u \\ u(0) &= \underline{u}\end{aligned}$$

- Because θ is pinned down without imposing steady state, these transition rates are constant even away from the steady state
- In the steady state, $\dot{u} = 0$ and we have the Beveridge curve

$$u = \frac{\sigma}{\sigma + \lambda_u(\theta)}$$

Equilibrium characterization

- The steady-state equilibrium $\{w, \theta, u, v\}$ is characterized by

$$\text{JC:} \quad w = y - \frac{c(r + \sigma)}{\lambda_v(\theta)}$$

$$\text{WC:} \quad w = (1 - \gamma)b + \gamma(y + c\theta)$$

$$\text{BC:} \quad u = \frac{\sigma}{\sigma + \lambda_u(\theta)}$$

$$\text{TD:} \quad v = \theta u$$

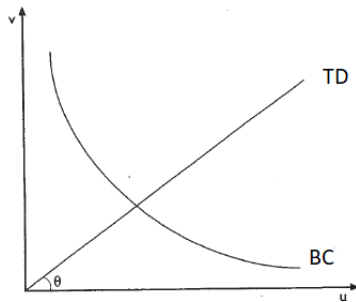
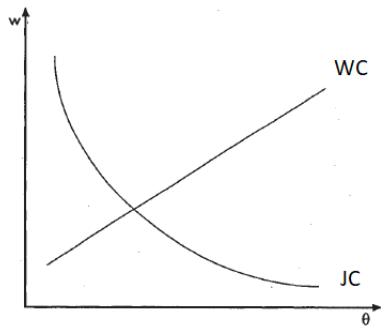
- The system is *block recursive*: solve first for θ using JC and WC

$$\frac{c}{\lambda_v(\theta)} = \frac{(1 - \gamma)(y - b)}{r + \sigma + \lambda_u(\theta)\gamma}$$

- ▶ LHS: firm cost of posting a vacancy
- ▶ RHS: firm value of a match

- Then recover u and v from BC and TD
- This decomposition is useful both analytically and graphically

Graphical equilibrium



- Once JC and WC are in hand, comparative statics are straightforward:

$$\text{JC:} \quad w = y - \frac{c(r + \sigma)}{\lambda_v(\theta)}$$

$$\text{WC:} \quad w = (1 - \gamma)b + \gamma(y + c\theta)$$

$$\text{BC:} \quad u = \frac{\sigma}{\sigma + \lambda_u(\theta)}$$

$$\text{TD:} \quad v = \theta u$$

- $b \uparrow \Rightarrow w \uparrow$ and $\theta \downarrow \Rightarrow u \uparrow$ and $v \downarrow$
 - Intuition? How does this compare to McCall model?
- $\gamma \uparrow \Rightarrow w \uparrow$ and $\theta \downarrow \Rightarrow u \uparrow$ and $v \downarrow$
 - Intuition?
- $y \uparrow \Rightarrow w \uparrow$ and $\theta \uparrow \Rightarrow u \downarrow$ and $v \uparrow$
 - Intuition?

Application: unemployment benefit extensions

- Did the unemployment benefit extensions in the US during the Great Recession contribute to the sharp increase in the unemployment rate?
- In lecture 7, we saw that a rich micro literature has estimated the partial equilibrium effect of extending unemployment benefits on unemployment duration
 - ▶ Estimates, however, not large enough to explain recession spike (Chetty's rule of thumb: 10 weeks of extra UI gives 1 week of extra unemployment)
- DMP emphasizes a different **general equilibrium channel** through which unemployment benefits affect unemployment: **vacancy creation**
- This motivates empirical work that tries to estimate the general-equilibrium effect

- US Great Recession: the federal benefit extension program depended on state-level characteristics, such as the increase in the state-level unemployment rate
- Chodorow-Reich-Coglienesse-Karabarbounis (QJE 2018) exploit state-level differences in duration extension due to real-time mismeasurement of unemployment rate: find no effect
- Hagedorn-Karahan-Manovskii-Mitman (2026) exploit policy discontinuity at US state borders: find big effect
- See also Marinescu (JPubE 2017) and Fredriksson-Söderström (JPubE 2020). The debate is ongoing.

Endogenous separations

Extension: endogenous separations

- How do we endogenize the separation decision?
- Hypothesis: Separations occur when match surplus becomes negative
- Consider the following twist to our model:
 - ▶ Production is xy , where y is aggregate productivity, and x is match-specific productivity
 - ▶ When forming a match, $x = 1$
 - ▶ A matched firm-worker pair draws match-specific productivity shocks $x \sim \Gamma(x)$, $x \in [0, 1]$, at arrival rate λ_x
 - ▶ If $J(x) < V$, the match is terminated
 - ▶ In each period, the firm-worker pair renegotiates the wage according to Nash bargaining
- We focus on the steady state
- More generally, we might also require $W(x) < U$ for separation, but let us start with the simpler case in which only the firm decides

Reservation productivity

- Firm Bellman equations:

$$rJ(x) = xy - w(x) + \lambda_x \left[\int_0^1 \max\{V, J(\epsilon)\} d\Gamma(\epsilon) - J(x) \right]$$

$$rV = -c + \lambda_v(\theta)(J(1) - V)$$

- Suppose $J(x)$ is increasing in x (this will be proved later)
- Then there is a reservation productivity x_R below which matches are dissolved, satisfying $J(x_R) = V$
- Rewriting the Bellman equation for job values gives

$$rJ(x) = xy - w(x) + \lambda_x \left[\int_0^1 \max\{V, J(x')\} d\Gamma(x') - J(x) \right]$$

$$\Rightarrow rJ(x) = xy - w(x) + \lambda_x \left[\int_{x_R}^1 (J(x') - J(x)) d\Gamma(x') + \int_0^{x_R} (V - J(x)) d\Gamma(\epsilon) \right]$$

$$\Leftrightarrow rJ(x) = xy - w(x) + \lambda_x \left[\int_{x_R}^1 (J(x') - J(x)) d\Gamma(x') + (V - J(x))\Gamma(x_R) \right]$$

- Taking x_R as given, the worker Bellman equations are

$$\begin{aligned}rW(x) &= w(x) + \lambda_x \left[\int_{x_R}^1 (W(x') - W(x)) d\Gamma(x') + (U - W(x)) \Gamma(x_R) \right] \\rU &= b + \lambda_u(\theta) (W(1) - U)\end{aligned}$$

Nash bargaining with rebargaining

- With Nash rebargaining each time a match-specific shock arrives, we have

$$\gamma(J(x) - V) = (1 - \gamma)(W(x) - U)$$

or

$$\begin{aligned}W(x) - U &= \gamma S(x) \\J(x) - V &= (1 - \gamma)S(x)\end{aligned}$$

- Implication: $J(x) - V < 0 \Rightarrow W(x) - U < 0$, so terminations are mutually beneficial
 - ▶ This is only true because of the rebargaining assumption (“flexible wage setting”)

Equilibrium with endogenous separations

- An equilibrium is a collection $\{W, U, J, V, w, x_R, \theta\}$ satisfying

- ▶ **Bellman equations:**

$$\begin{aligned}rW(x) &= w(x) + \lambda_x \left[\int_{x_R}^1 (W(x') - W(x)) d\Gamma(x') + (U - W(x)) \Gamma(x_R) \right] \\rU &= b + \lambda_u(\theta) (W(1) - U) \\rJ(x) &= xy - w(x) + \lambda_x \left[\int_{x_R}^1 (J(x') - J(x)) d\Gamma(x') + (V - J(x)) \Gamma(x_R) \right] \\rV &= -c + \lambda_v(\theta) (J(1) - V)\end{aligned}$$

- ▶ **Separation decision:**

$$J(x_R) = V$$

- ▶ **Free entry:**

$$V = 0$$

- ▶ **Wage setting rule:**

$$\gamma(J(x) - V) = (1 - \gamma)(W(x) - U)$$

- Seven equations, seven unknowns

- Given θ and x_R , unemployment and vacancies follow from

$$\begin{aligned}\dot{u} &= \sigma(x_R)(1-u) - \lambda_u(\theta)u \\ u(0) &= \underline{u} \\ \theta &= \frac{v}{u}\end{aligned}$$

where $\sigma(x_R) = \lambda_x \Gamma(x_R)$

- In steady state:

$$u = \frac{\lambda_x \Gamma(x_R)}{\lambda_x \Gamma(x_R) + \lambda_u(\theta)}$$

- Combine the Bellman equations, the surplus-splitting rule, and the free-entry condition $V = 0$ to find the wage curve:

$$w(x) = (1 - \gamma)b + \gamma(xy + c\theta)$$

Match surplus

- Use the surplus-splitting rule, the wage curve, and the free-entry condition to derive the Bellman equation for total match surplus $S(x)$ (**DO AT HOME!**):

$$rS(x) = xy - b - \frac{\gamma}{1-\gamma}c\theta + \lambda_x \left[\int_{x_R}^1 S(x')d\Gamma(x') - S(x) \right]$$

- Evaluate at $x = x_R$:

$$0 = x_R y - b - \frac{\gamma}{1-\gamma}c\theta + \lambda_x \left[\int_{x_R}^1 S(x')d\Gamma(x') \right]$$

- Subtract the equation at x_R to get

$$\begin{aligned} rS(x) &= rS(x) - rS(x_R) \\ &= (x - x_R)y - \lambda_x S(x) \end{aligned}$$

or

$$S(x) = \frac{y(x - x_R)}{r + \lambda_x}$$

- Interpretation: surplus rises linearly with match quality above the reservation threshold

- Nash bargaining:

$$\begin{aligned} J(x) - V &= (1 - \gamma)S(x) \\ &= \frac{(1 - \gamma)y(x - x_R)}{r + \lambda_x} \end{aligned}$$

- Free entry in Bellman for vacancy value:

$$J(1) - V = \frac{c}{\lambda_v(\theta)}$$

- Together:

$$\frac{c}{\lambda_v(\theta)} = (1 - \gamma)y \frac{1 - x_R}{r + \lambda_x}$$

- This is the *job-creation curve* in θ, x_R -space

Job-destruction curve

- Bellman equation for surplus:

$$rS(x) = xy - b - \frac{\gamma}{1-\gamma}c\theta + \lambda_x \left[\int_{x_R}^1 S(x')d\Gamma(x') - S(x) \right]$$

- Evaluated at $x = x_R$:

$$0 = x_R y - b - \frac{\gamma}{1-\gamma}c\theta + \lambda_x \left[\int_{x_R}^1 S(x')d\Gamma(x') \right]$$

- Using our solution for $S(x)$:

$$0 = x_R y - b - \frac{\gamma}{1-\gamma}c\theta + \lambda_x \left[\int_{x_R}^1 \frac{y(x - x_R)}{r + \lambda_x} d\Gamma(x') \right]$$

or

$$x_R y = b + \frac{\gamma}{1-\gamma}c\theta - y \frac{\lambda_x}{r + \lambda_x} \int_{x_R}^1 (x' - x_R) d\Gamma(x')$$

- This is the *job-destruction curve* in θ, x_R -space

- The pair $\{x_R, \theta\}$ is pinned down by the job-creation and job-destruction curves:

$$\frac{c}{\lambda_v(\theta)} = (1 - \gamma)y \frac{1 - x_R}{r + \lambda_x}$$
$$x_R y = b + \frac{\gamma}{1 - \gamma} c \theta - y \frac{\lambda_x}{r + \lambda_x} \int_{x_R}^1 (x' - x_R) d\Gamma(x')$$

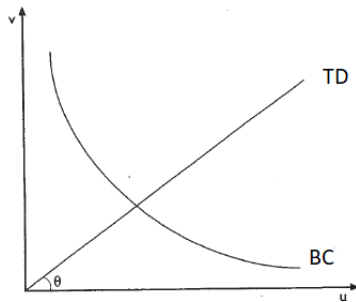
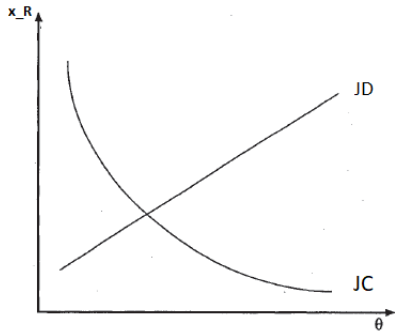
- Given x_R and θ , recover $\{v, u\}$ from

$$v = \theta u$$
$$u = \frac{\lambda_x \Gamma(x_R)}{\lambda_x \Gamma(x_R) + \lambda_u(\frac{v}{u})}$$

- Match-specific wages are then given by

$$w(x) = (1 - \gamma)b + \gamma(xy + c\theta)$$

Graphical equilibrium



- The key new object is the job-destruction curve: shocks now move both vacancy creation and separation behavior
- See the problem set for worked comparative statics

- DMP is a general-equilibrium theory of unemployment
 - ▶ emphasizes vacancy creation and job destruction as key mechanisms
- Core ingredients: matching, wage setting, and free entry
- With endogenous separations, the model also explains why low-productivity matches break up
- Today we focused on steady-state equilibrium and comparative statics
- Next lecture: welfare and dynamics